

Expanded Search for Toroidal Isomers

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Preliminary Exam / Masters Defense

2026-05-13

Title

Liquid Drop Model (1970s) - Original Theory

Spherical Nucleus Binding Energy

$$E_B = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(N-Z)^2}{A} \pm \delta(N, Z)$$

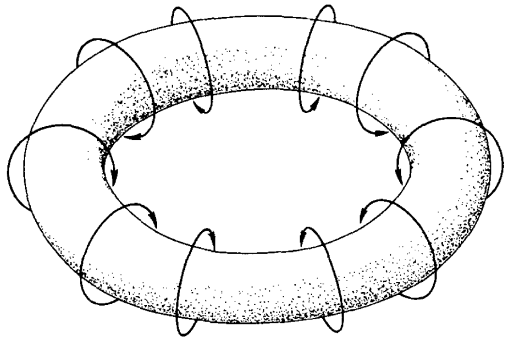
Liquid Drop Model (1970s) - Original Theory

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- E_B depends on **spherical** assumptions.

What happens under deformed Toroidal configurations at high angular momentum?



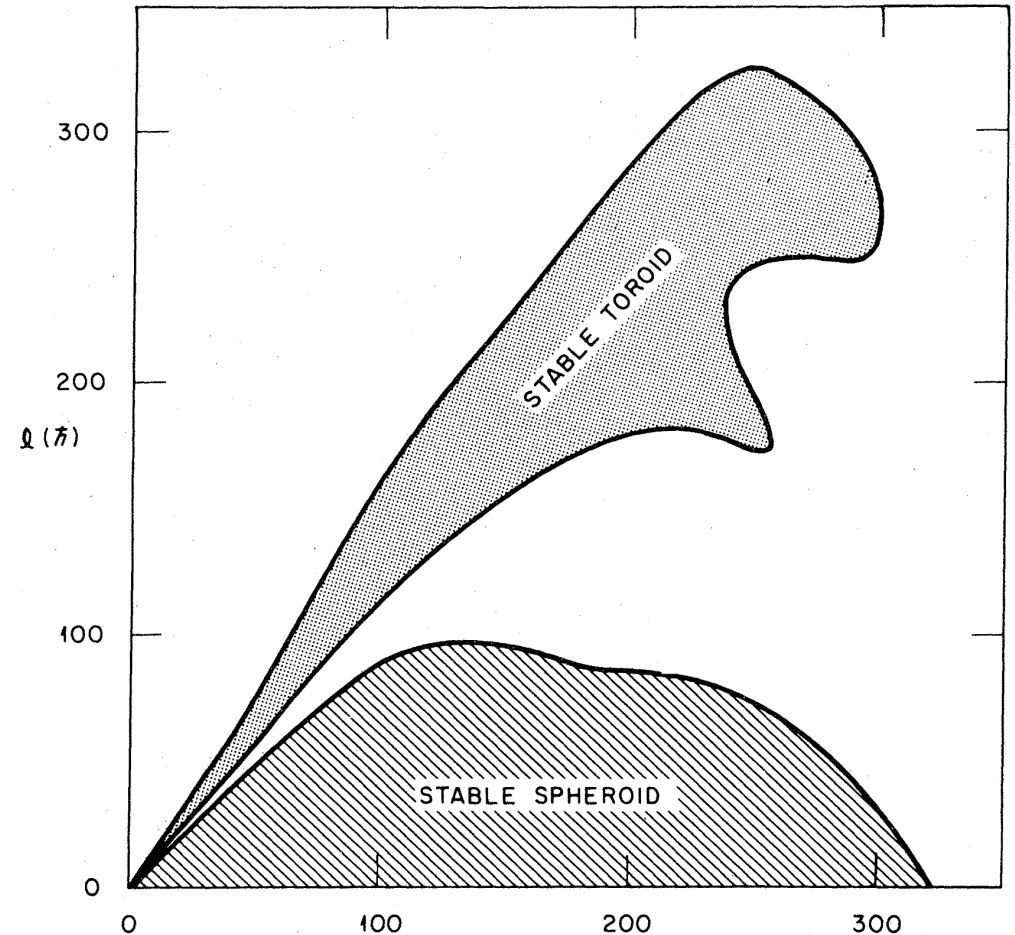
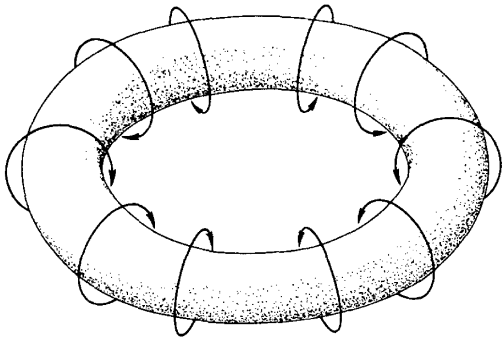
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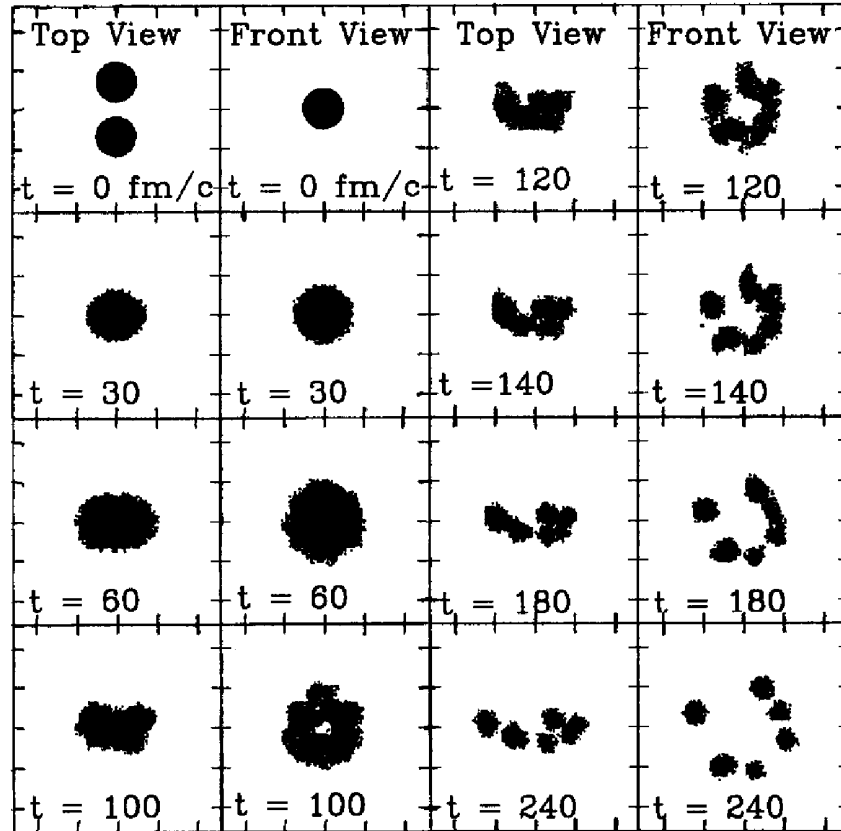
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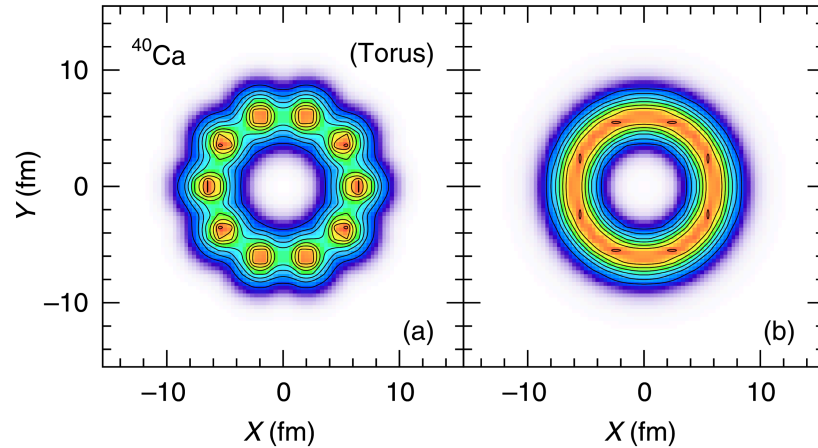
Boltzmann-Uehling-Uhlenbeck (BUU) (1990s) - Dynamic Formation

$^{92}\text{Mo} + ^{92}\text{Mo}$, $E/A = 100$ MeV, $b=0$, Stiff EOS



- Direct collisions produced outwardly expanding toroids.
- Toroids seem to be metastabilized
- Decay into symmetric particles
 - $r_{\text{particle}} \approx d_{\text{toroid}}$

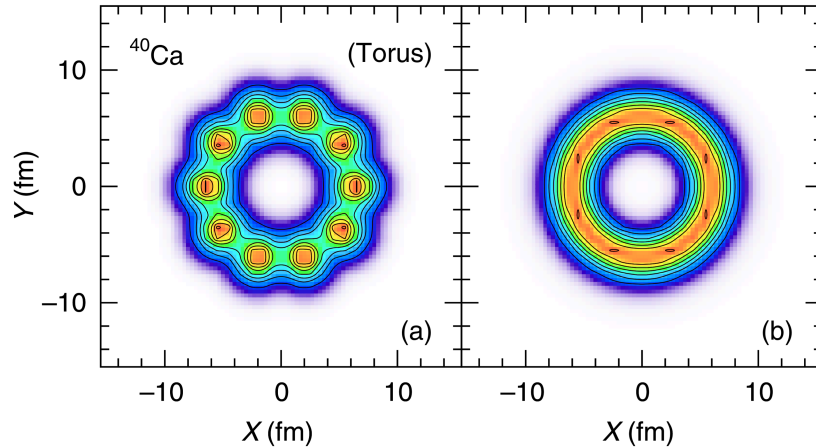
Cranked Skyrme Hartree Fock (cSHF) (2010s) - Specific Energy Predictions



PRL **109**, 232503 (2012)

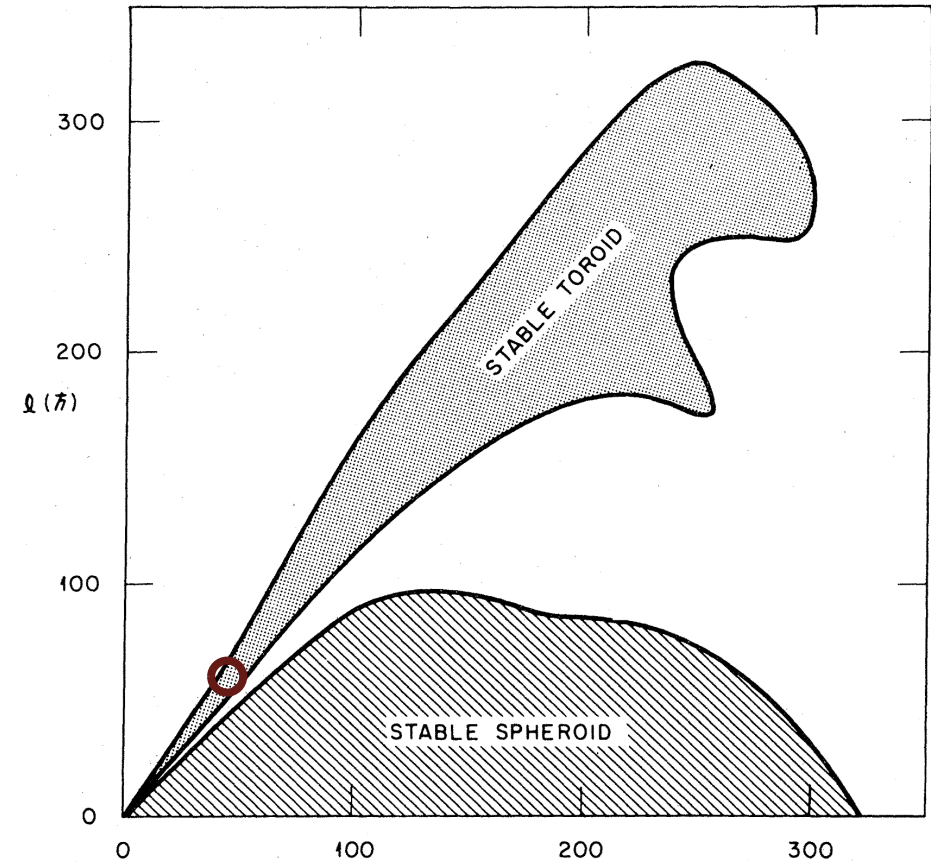
- Predicted state in ^{40}Ca
 - $J_z = 60\hbar$
 - $E^* \approx 175 \text{ MeV}$
- Angular momentum and mass consistent with Wong's LDM

Cranked Skyrme Hartree Fock (cSHF) (2010s) - Specific Energy Predictions



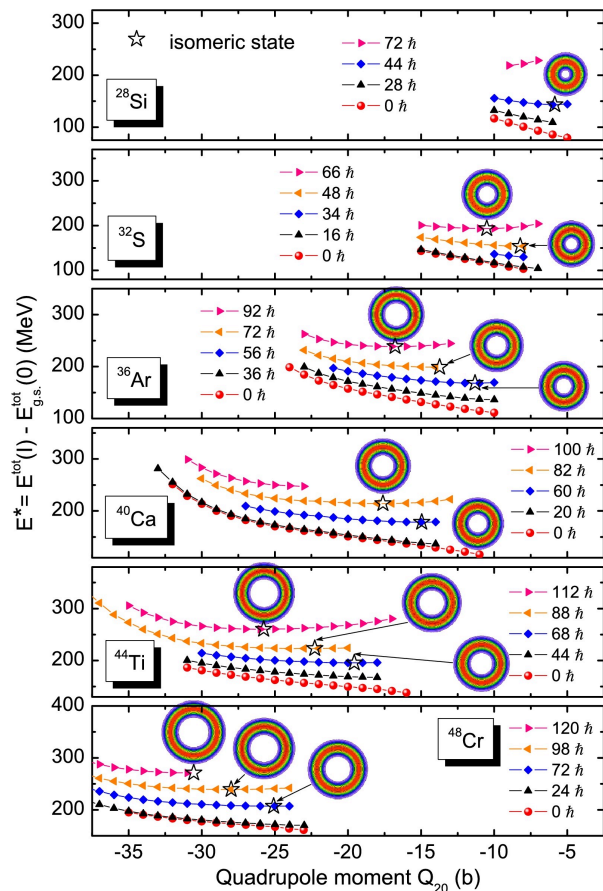
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PLB **738**, 401-404 (2014)

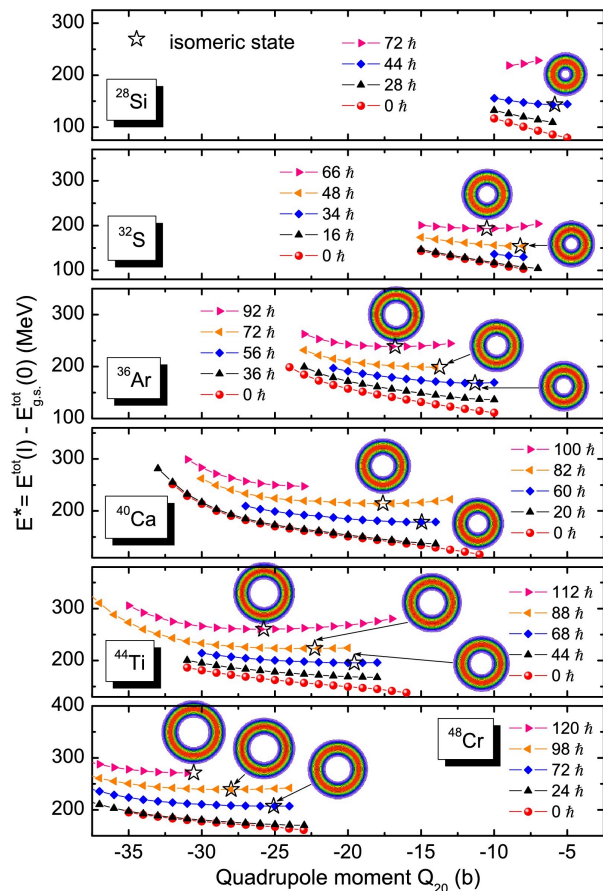


A. Staszczak and C.-Y. Wong predict 18 Toroidal Isomers

	E^* [MeV]	I [$\hbar$]	d [fm]	ρ_{max}/ρ_0
^{28}Si	143.18	44	1.45	0.74
^{32}S	153.87	48	1.42	0.76
	193.35	66	1.40	0.67
^{36}Ar	168.03	56	1.40	0.78
	198.63	72	1.39	0.71
	238.56	92	1.37	0.64
^{40}Ca	178.36	60	1.40	0.79
	214.23	82	1.39	0.73
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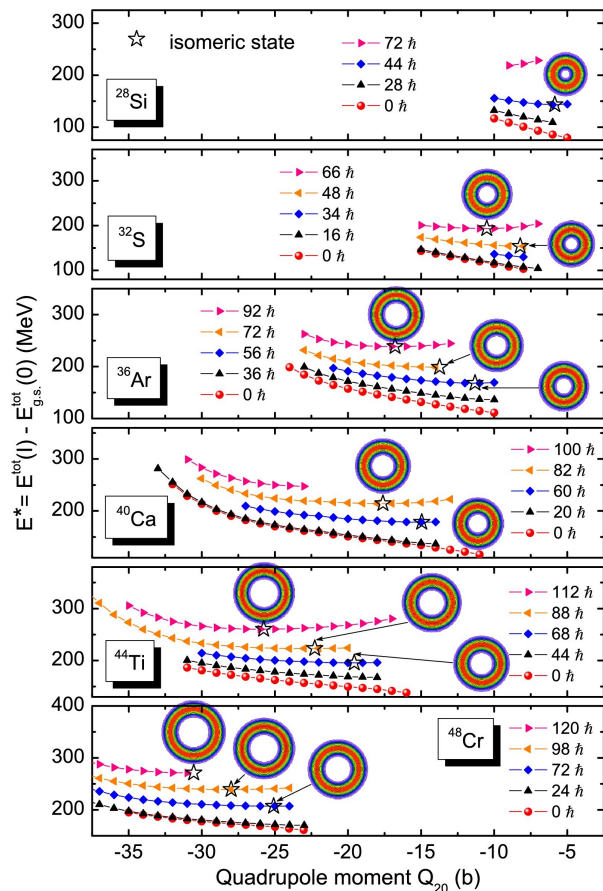
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Ichikawa's ^{40}Ca is reproduced

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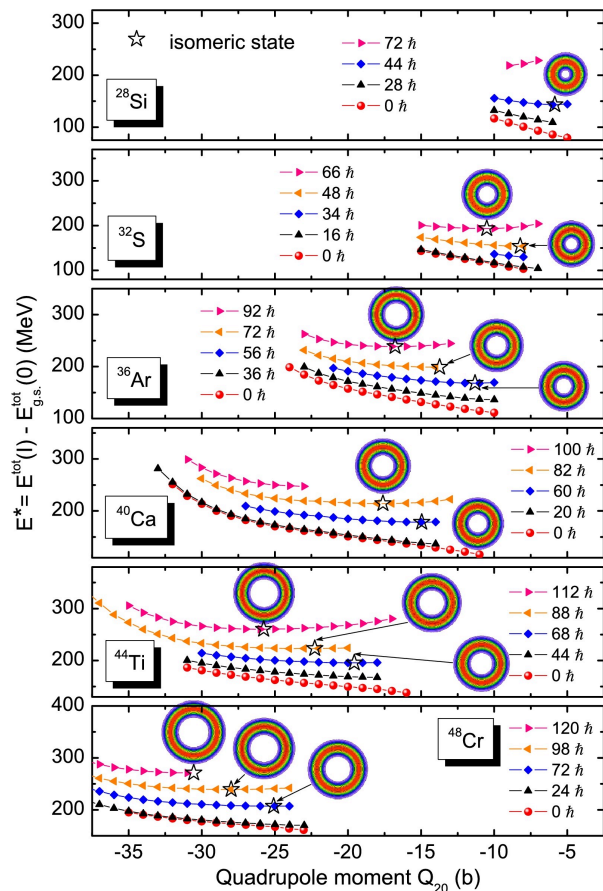
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Minor axis stays about constant ($d \approx 1.4$ fm)

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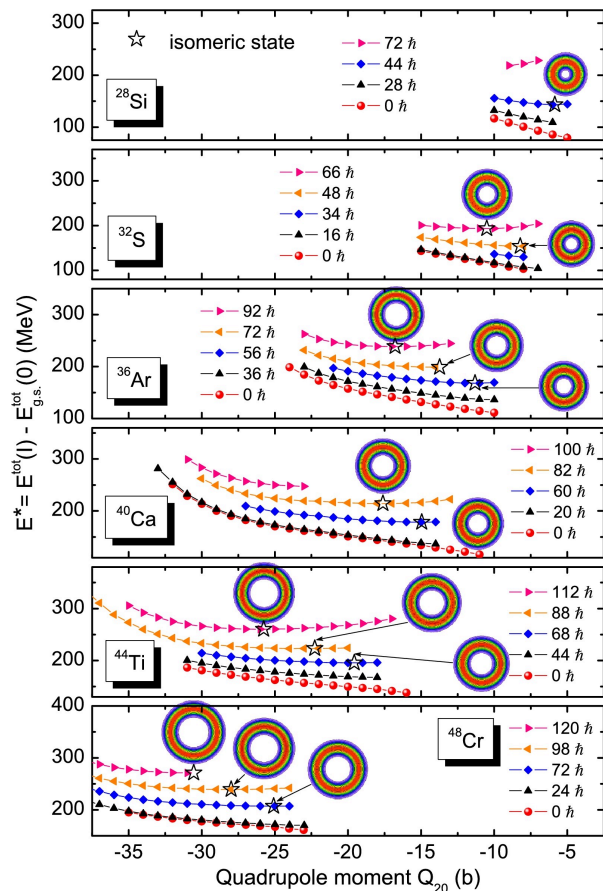
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Low density

Cranked Skyrme Hartree Fock (cSHF) (2010s) - Specific Energy Predictions

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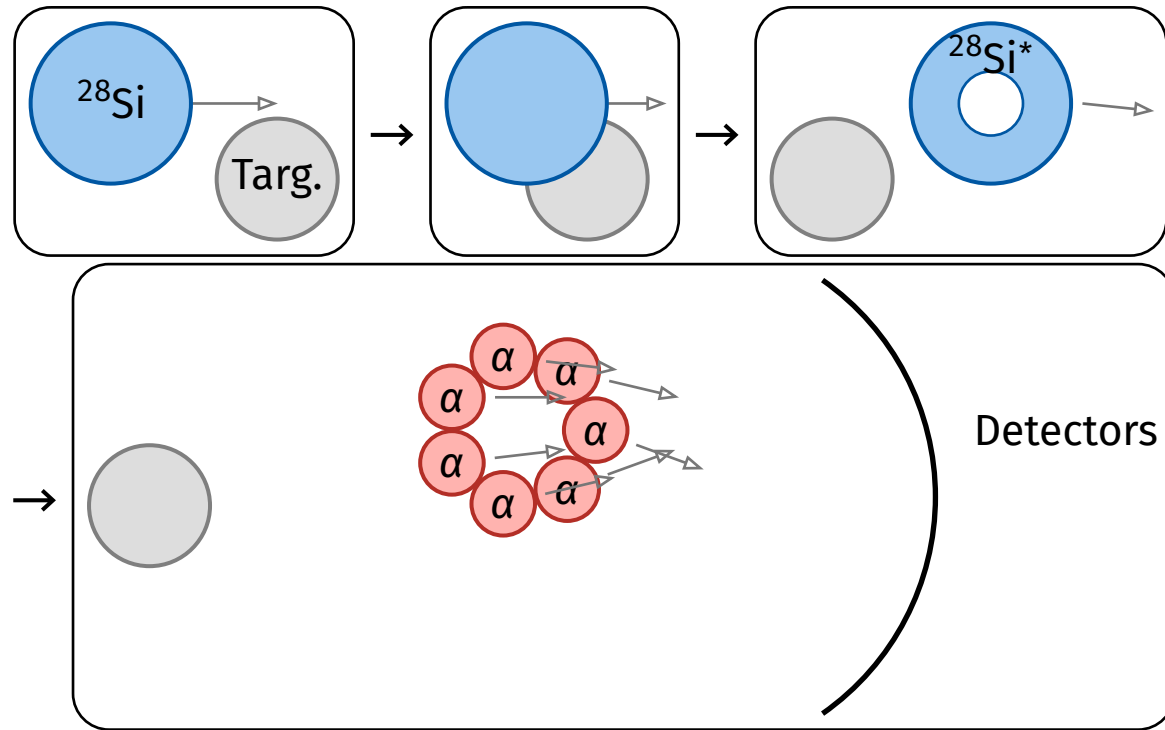


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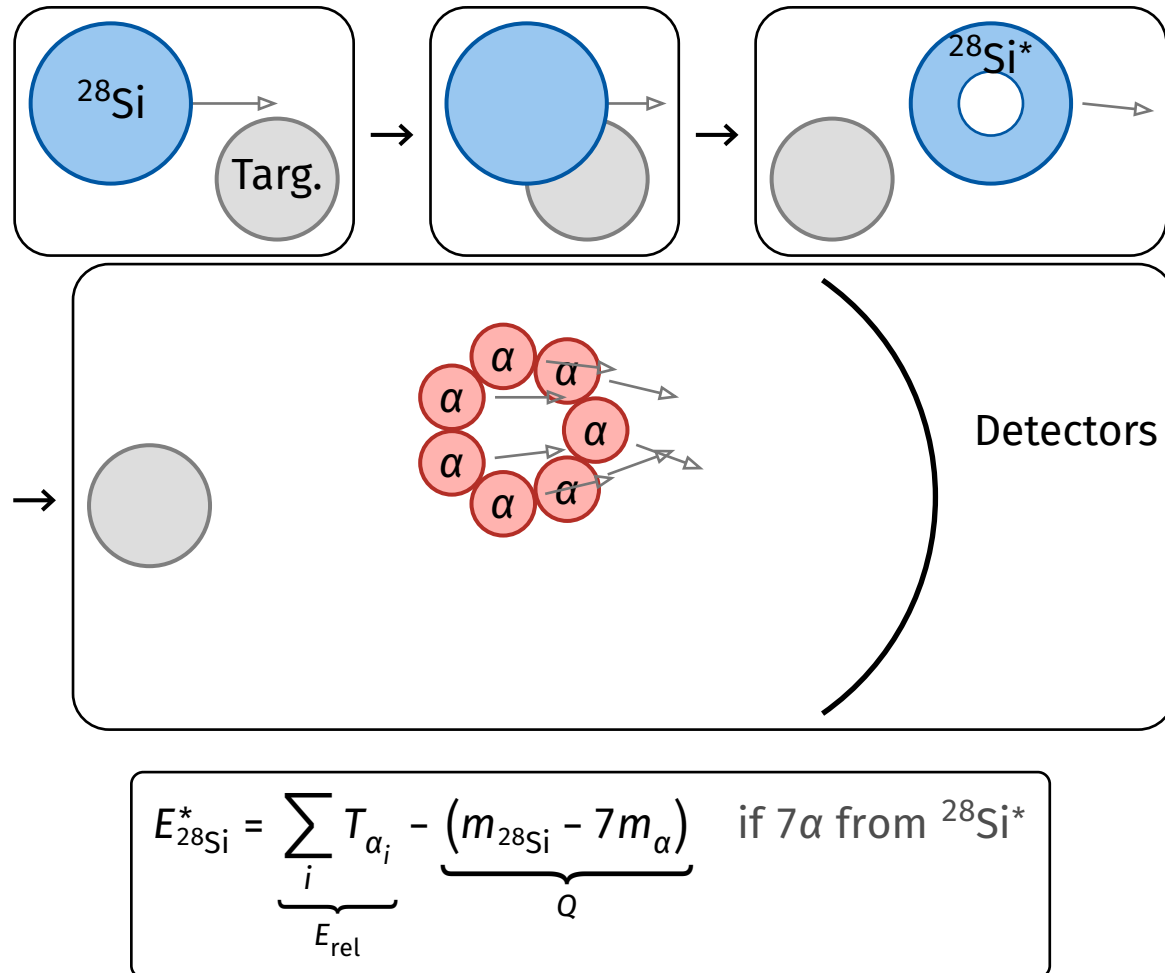
Several states to investigate, including $^{28}\text{Si}(E^* = 143 \text{ MeV})$

First Experimental Evidence (2019)

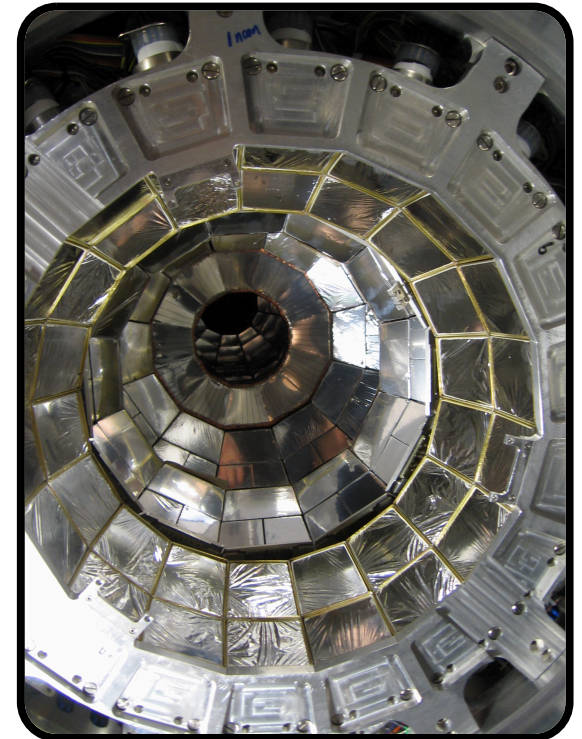


$$E_{^{28}\text{Si}}^* = \underbrace{\sum_i T_{\alpha_i}}_{E_{\text{rel}}} - \underbrace{(m_{^{28}\text{Si}} - 7m_{\alpha})}_Q \quad \text{if } 7\alpha \text{ from } ^{28}\text{Si}^*$$

First Experimental Evidence (2019)



Neutron Ion Multidetector For Reaction Oriented Dynamics (NIMROD)



First Experimental Evidence (2019)

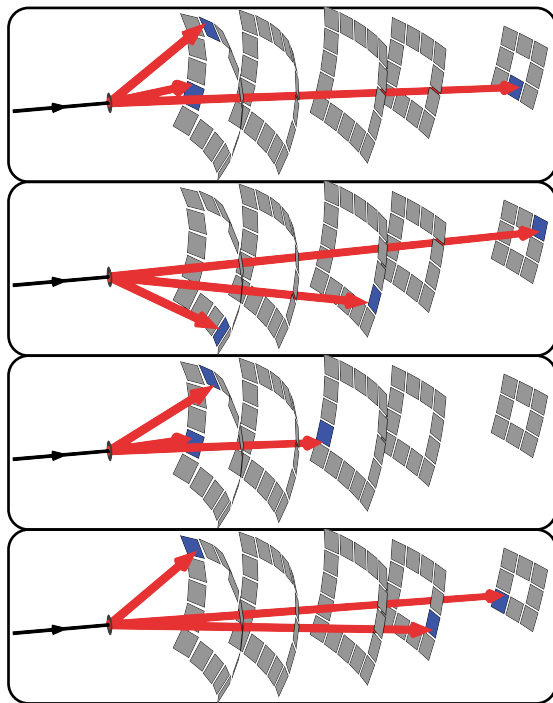
test1

test2

test3

Creating a Mixed Event

Real Events

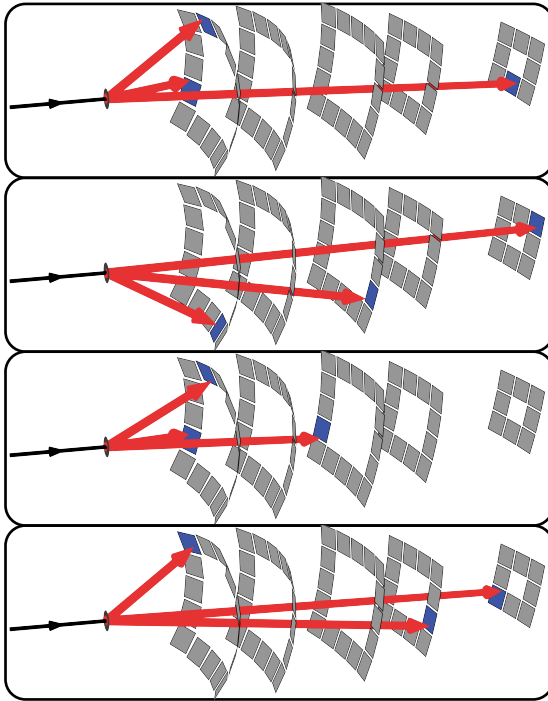


⋮

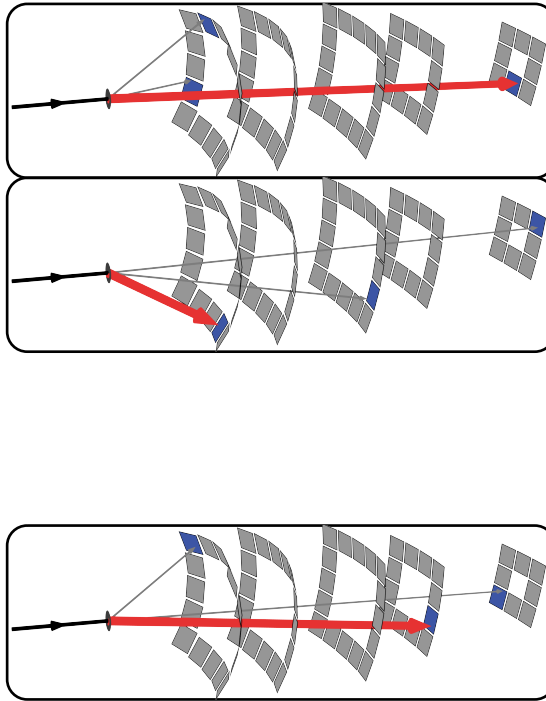
- Start w/ events containing all particles

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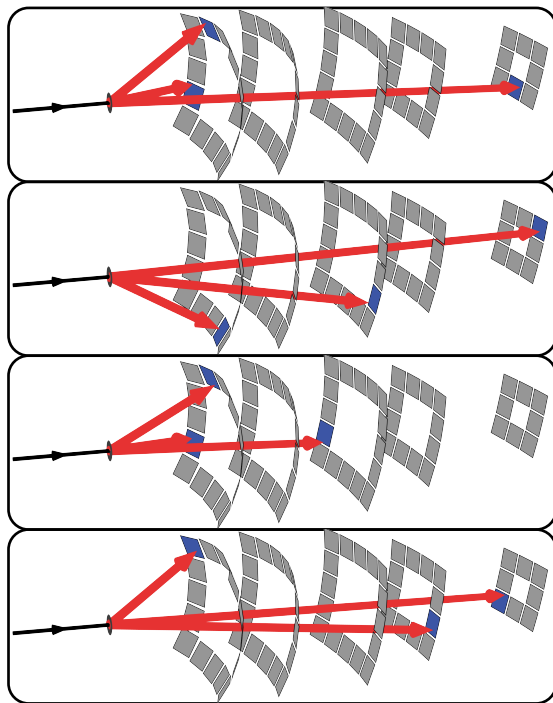
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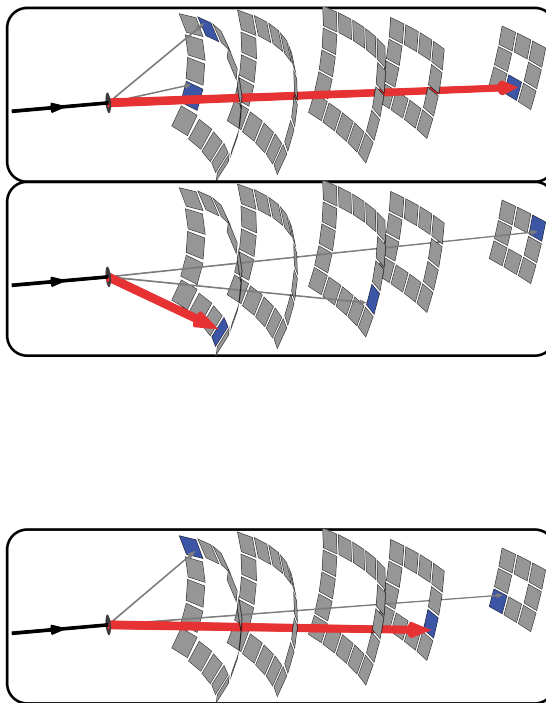
- Start w/ events containing all particles
- Select particles from separate events

Creating a Mixed Event

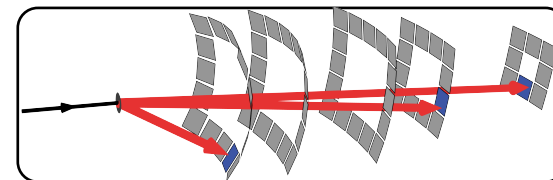
Real Events



⋮



Mixed Events



⋮

- Start w/ events containing all particles
- Select particles from separate events
- Construct new event

Outline

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